

64-Bit SRAM Array Design with Custom Peripheral Circuitry (45 nm PDK)

Course: ESE 5700 - Digital Integrated Circuits and VLSI

Institution: University of Pennsylvania

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Project Overview

This project implements a complete 64-bit (16×4) Static Random-Access Memory (SRAM) array using standard 6-transistor (6T) memory cells and custom-designed peripheral circuitry. Two designs are developed and compared:

1. Baseline Design: Conventional full-VDD bitline precharge with tri-state output buffers
2. Optimized Design: VDD/2 bitline precharge with OTA-based differential sense amplifiers

The optimized design achieves a 3× improvement in Figure of Merit ($\text{FOM} = \text{Area} \times \text{Power} \times \text{Delay}$), demonstrating the impact of co-optimizing memory cells and peripheral circuits.

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1. Architecture

System-Level Overview

The SRAM array consists of:

- Memory Array: 16 rows $\times$ 4 columns of 6T SRAM cells (64 bits total)
- Shared Peripheral Circuits:
 - Non-overlapping clock generator
 - 4-to-16 dynamic row decoder
 - Clocked transmission gate control logic
- Per-Column Circuits:
 - Bitline precharge modules
 - Bitline equalizers
 - Write drivers
 - Read buffers/sense amplifiers

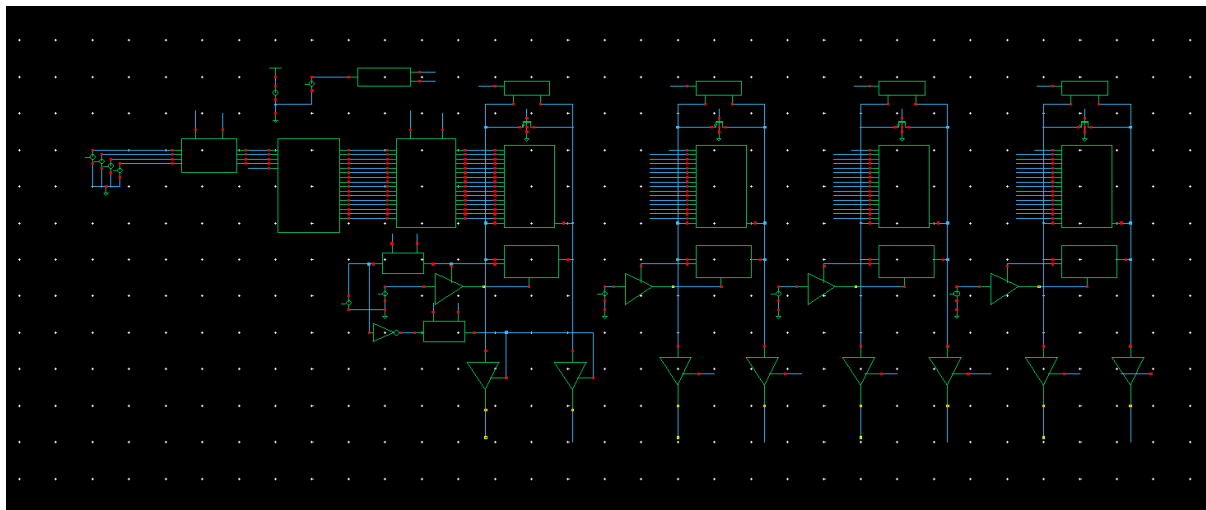


Figure 1. Fully synchronous, 64 bit 6T SRAM array Complete with peripheral circuitry.

Memory Cell: 6T SRAM

The storage element uses a standard 6-transistor configuration:

- Two cross-coupled inverters (4 transistors) provide bistable storage
- Two access transistors (2 transistors) controlled by word line enable read/write operations

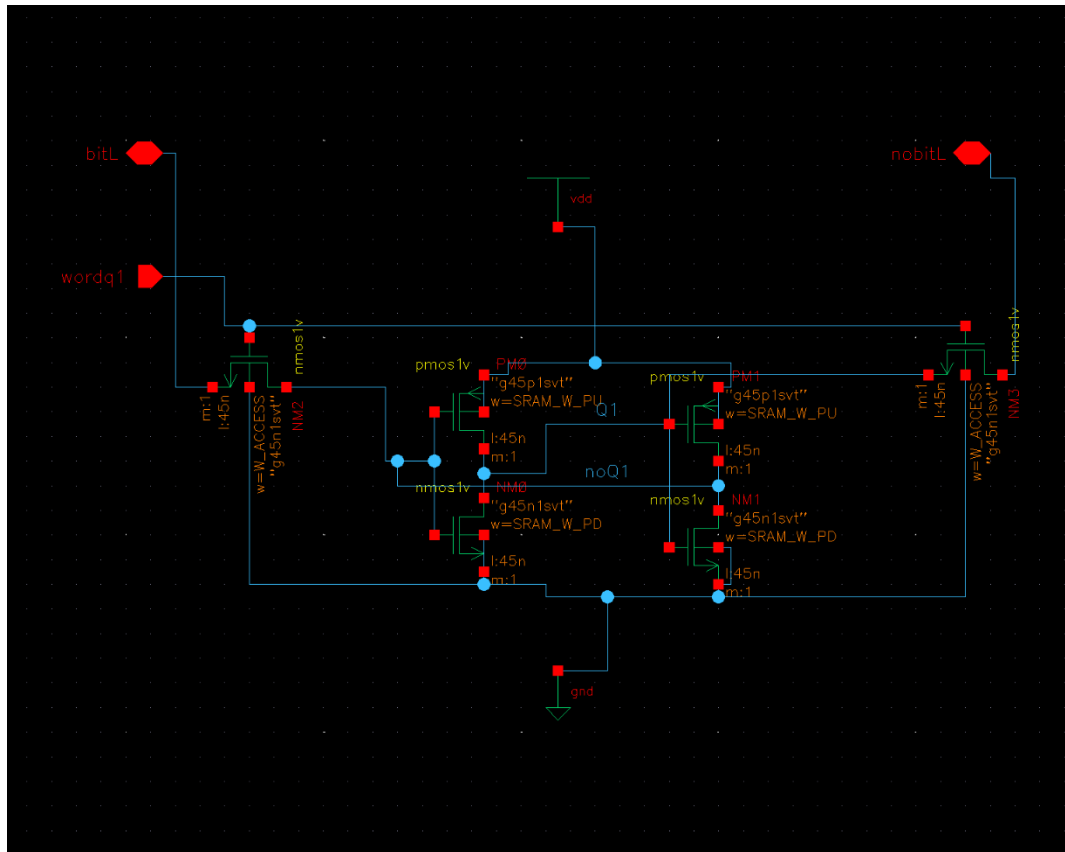


Figure 2. Standard 6T SRAM cell. Data is stored in the cross coupled inverter pair.

Key Design Challenges

1. Read Stability vs. Writeability Tradeoff:
 - Larger pull-down transistors improve read stability but hurt writeability
 - Smaller pull-up transistors improve writeability but reduce noise margins
2. Bitline Capacitance:
 - 16 cells share common bitlines, creating significant capacitive loading
 - Precharge strategy and sense amplifier design critically impact performance
3. Timing and Synchronization:
 - Non-overlapping clocks prevent race conditions and short-circuit currents
 - Proper sequencing of precharge, evaluation, and access phases is essential

2. Baseline Design

Architecture Details

1. Non-Overlapping Clock Generator

Converts a single input clock into two complementary, non-overlapping phases (CLK and CLKB) using cross-coupled NOR gates with buffered outputs. This ensures clean separation of precharge and active phases.

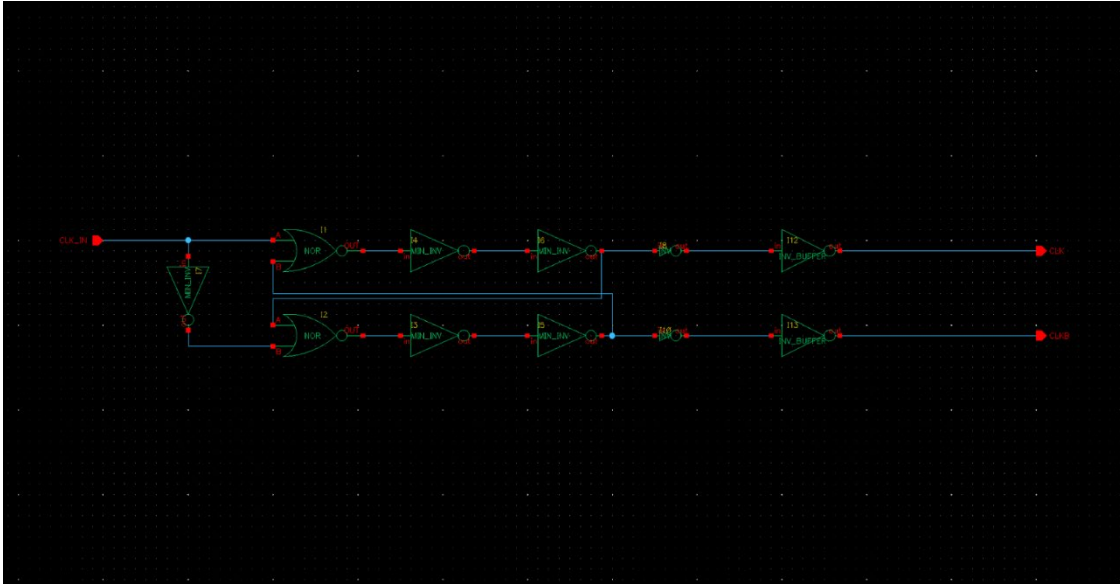


Figure 3. Schematic for non-overlapping clock generator. Used to synchronize all cell components.

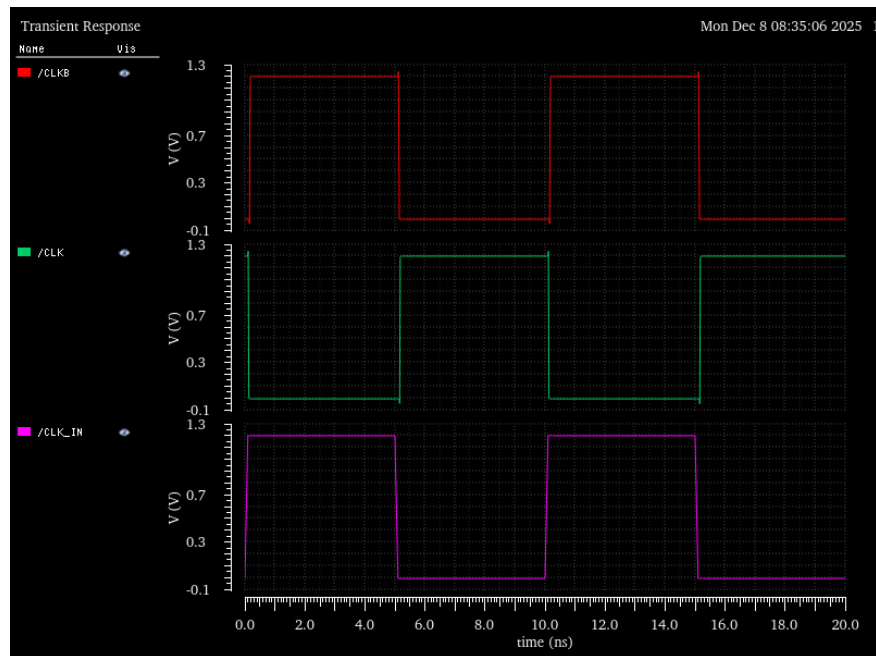


Figure 3. Test output waveforms. CLK and CLKB are clearly opposite in phase.

2. 4-to-16 Dynamic Decoder

Implements a precharged dynamic architecture for row selection:

- Precharges internal nodes during CLKB high
- Evaluates selected address during CLK high
- Only one word line activates per cycle

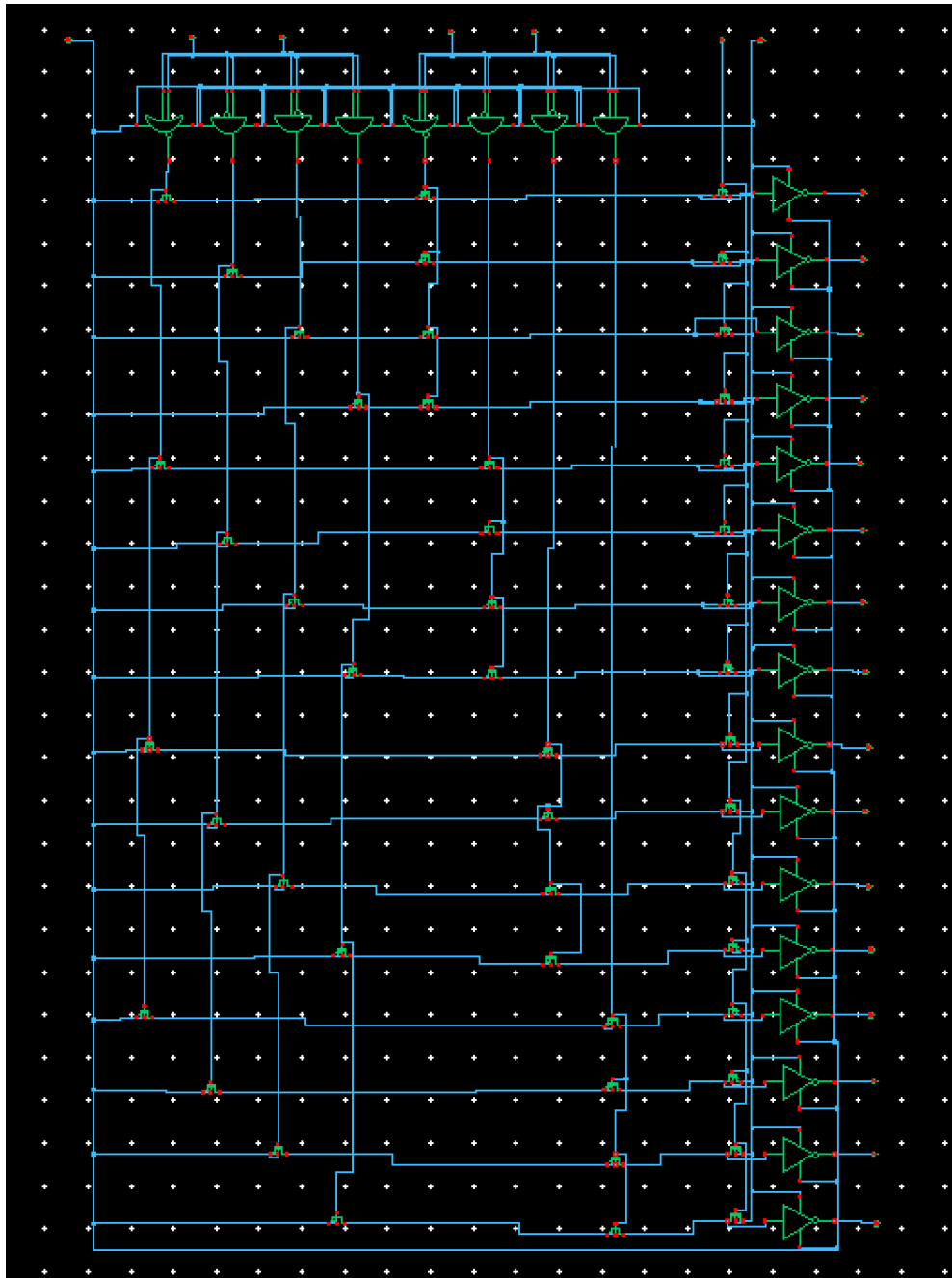


Figure 6. Full decoder architecture. The four bit input entered at the top of the decoder is transformed into the corresponding word line on the right side.

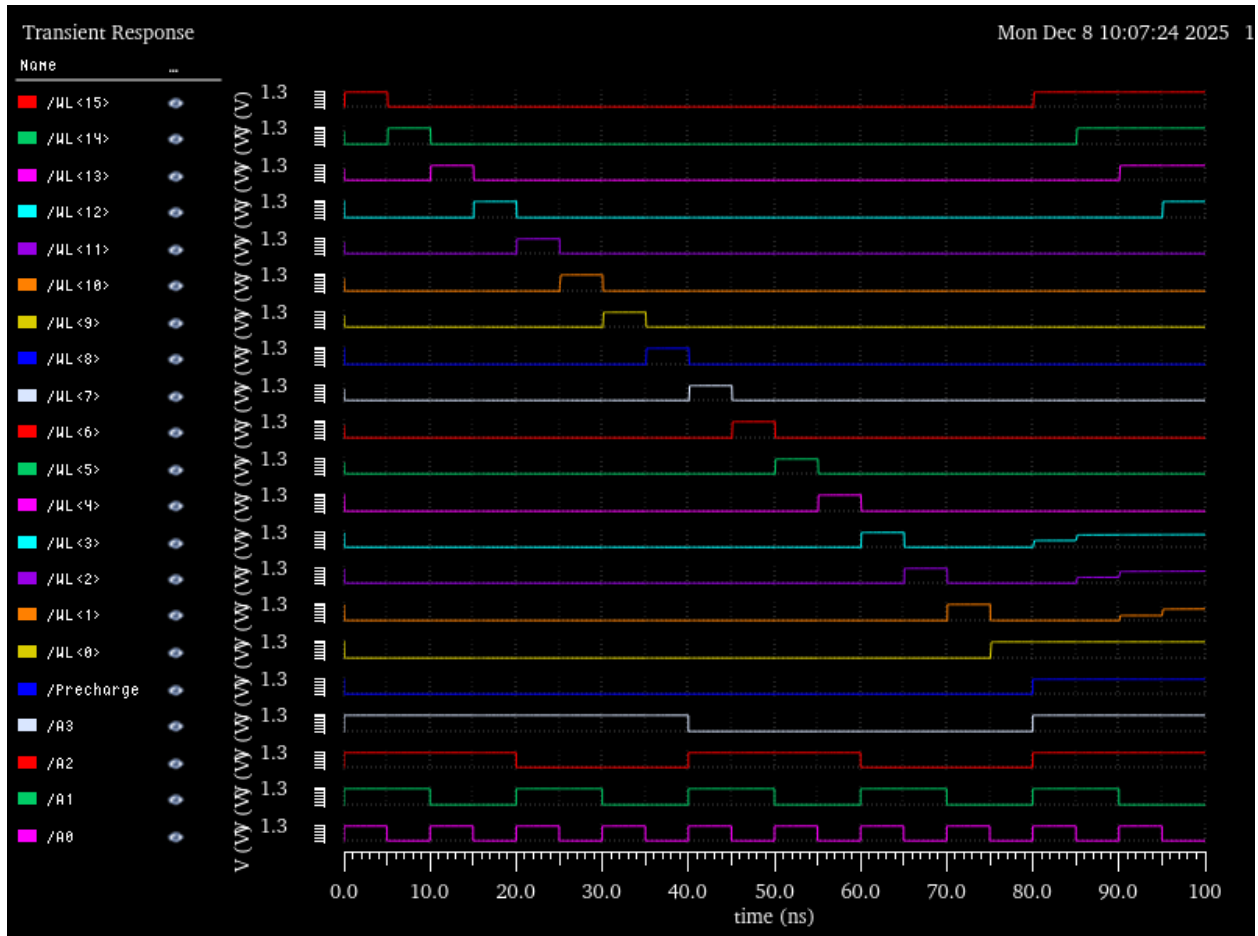


Figure 7. Test output waveforms. A<0:3> corresponds to the address bits, which are cycled through. WL<0:15> are the word lines that the decoder outputs to.

3. Full-VDD Bitline Precharge

PMOS pass transistors precharge both BL and BLB to VDD when CLK is low:

- Provides symmetric initial conditions for read/write
- Simple implementation with predictable behavior
- Higher dynamic power due to full-swing bitline transitions

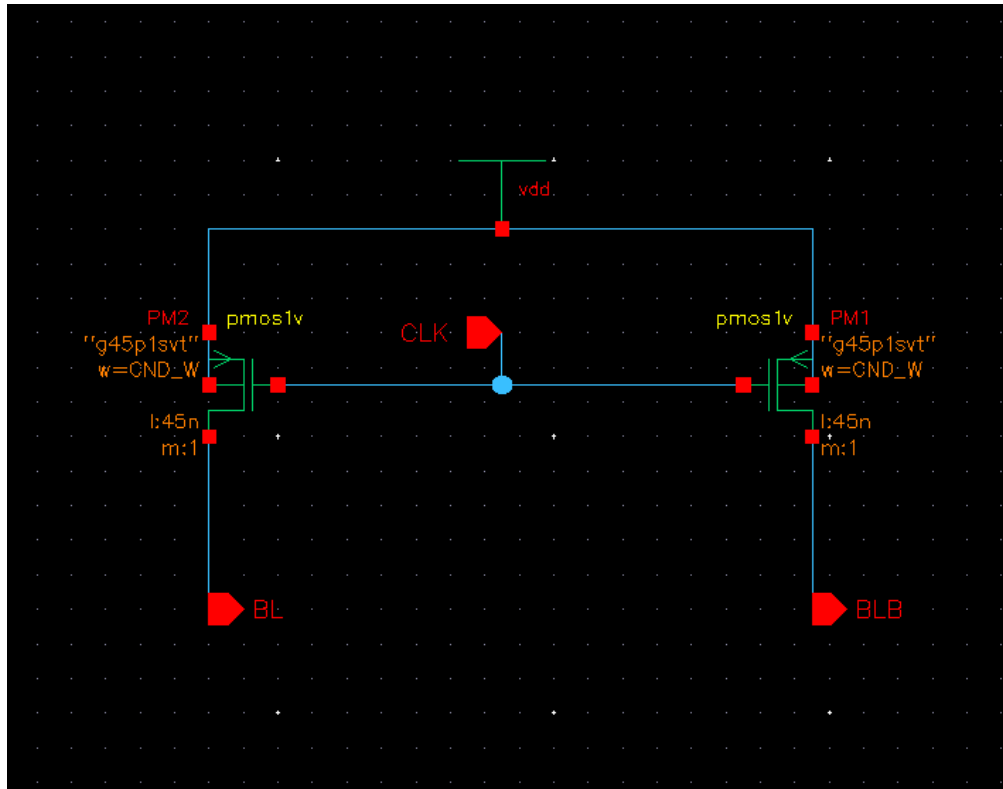


Figure 8. Full VDD precharge module schematic. PMOS devices turn on when the input clock signal goes low, effectively precharging the bitlines between word line drives.

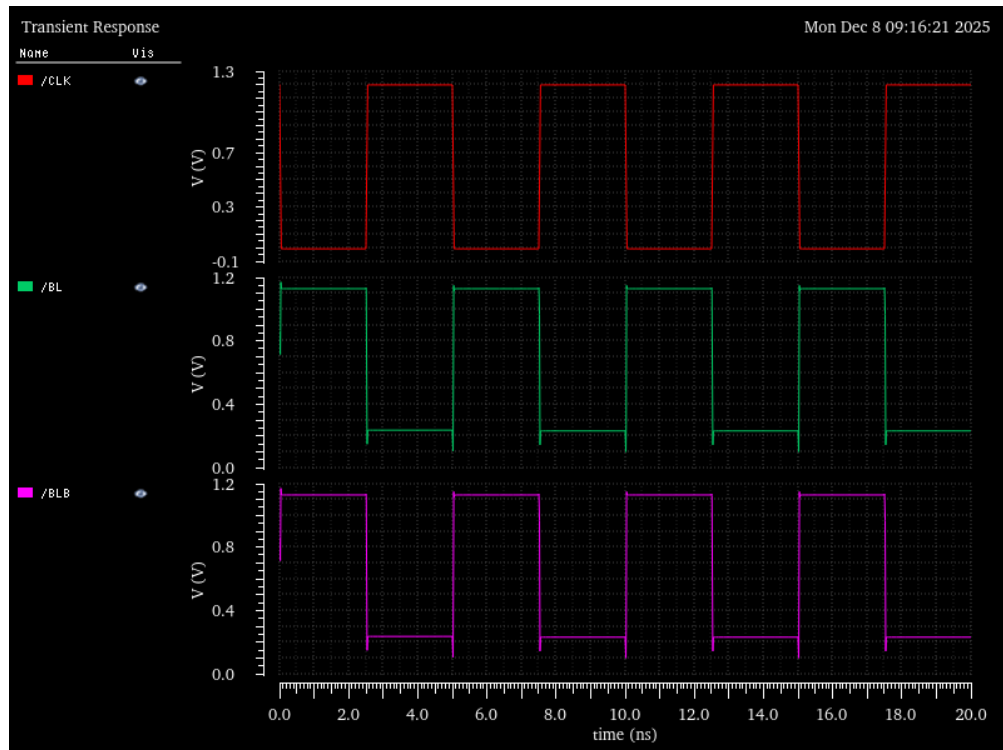


Figure 9. Test output waveforms. Whenever the clock signal is low, the precharge activates, bringing both BL and BLB to VDD.

4. Write Driver

Differential write driver controlled by Write Enable (WE):

- Data input drives one bitline through NMOS pass gate
- Inverted data drives complementary bitline
- Strong drive strength sized to overcome bitline capacitance
- Only active when WE is asserted

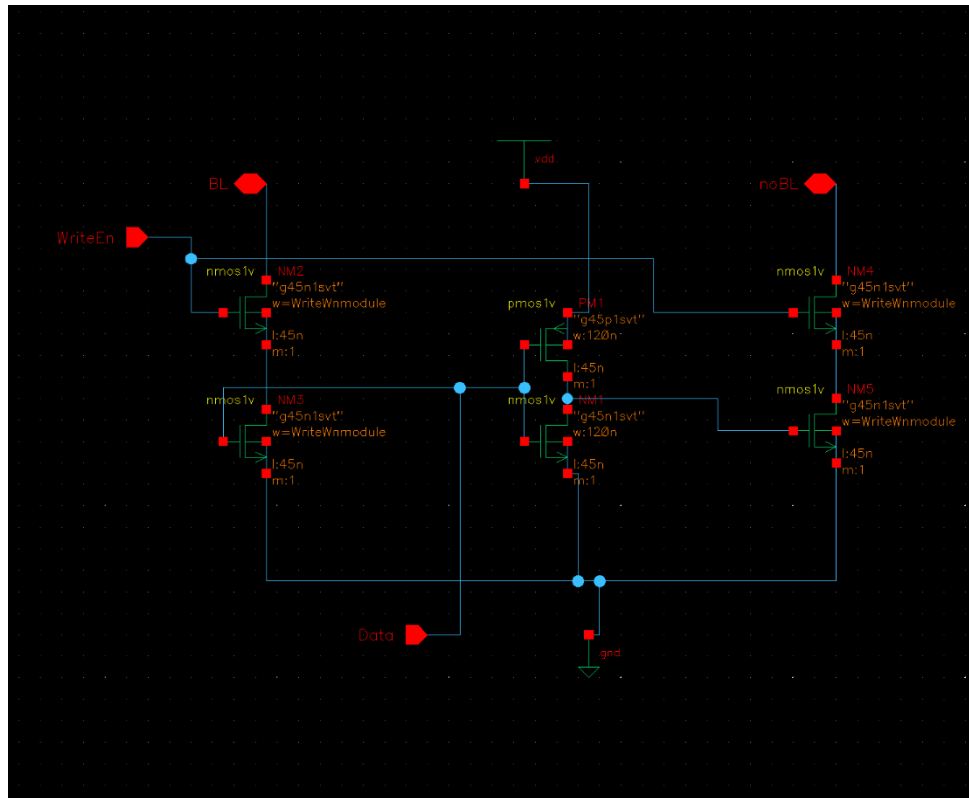


Figure 10. Write module schematic. Data is pushed to the bitline and the inverse of data is pushed to bitline bar. Write enable connects/disconnects the circuit from the bitlines.

5. Tri-State Read Buffer

Enables conditional readout from bitlines:

- Enable signal gates output stage
- When enabled, functions as standard buffer
- When disabled, output enters high-impedance state
- One buffer per column for independent readout

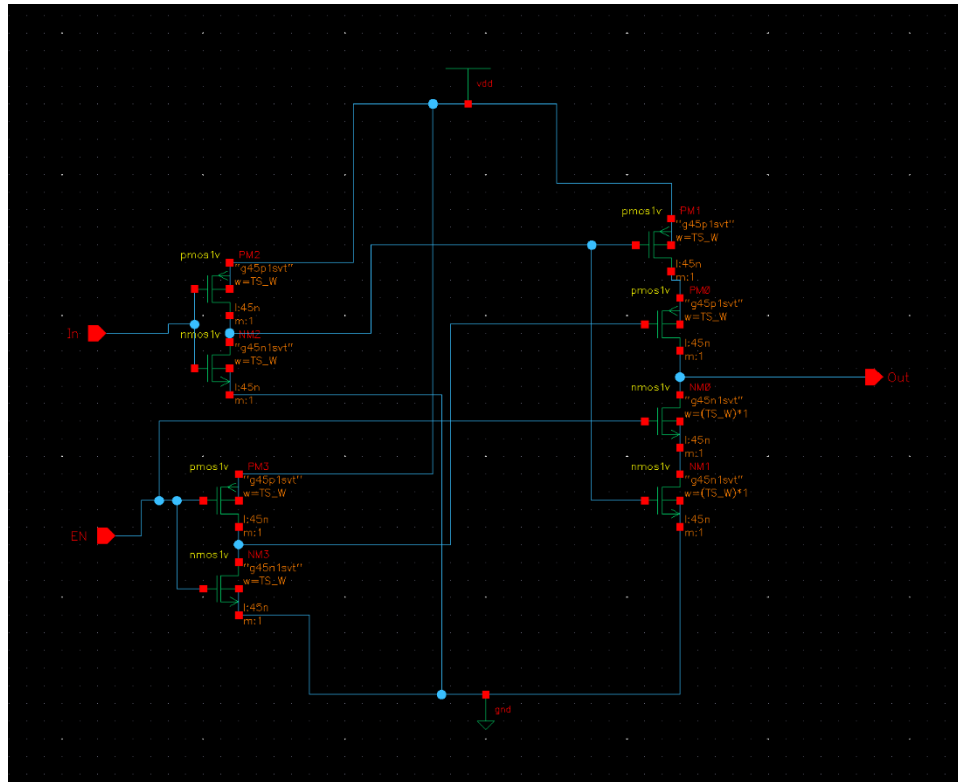


Figure 11. Tri state buffer module schematic. Output follows input while Enable is high. Otherwise, the output is a high impedance node.

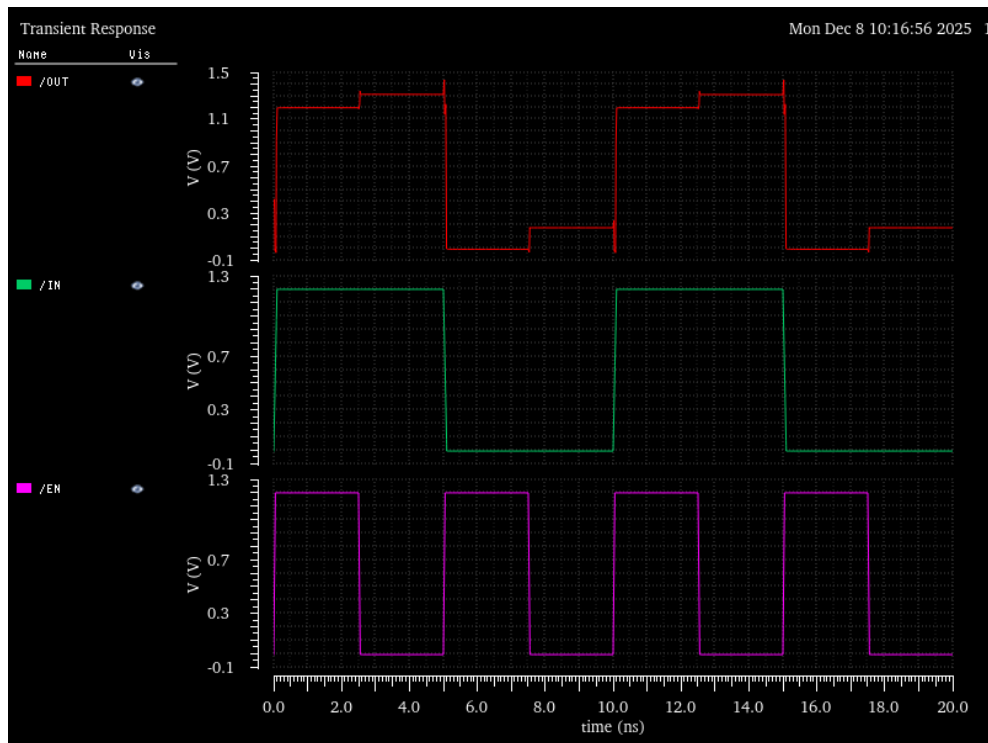


Figure 12. Test output waveform for tri state buffer. When EN is high, the output directly follows the input. When EN is low, the output is floating, which is represented by a drift to a higher voltage.

6. Clocked Transmission Gate with Pull-Down (TG-PD)

Ensures deterministic timing for control signals:

- Complementary pass transistors pass signals during CLK high
- Pull-down NMOS forces output to ground during CLKB high
- Prevents floating nodes and charge sharing
- Used for word line gating and control signal synchronization

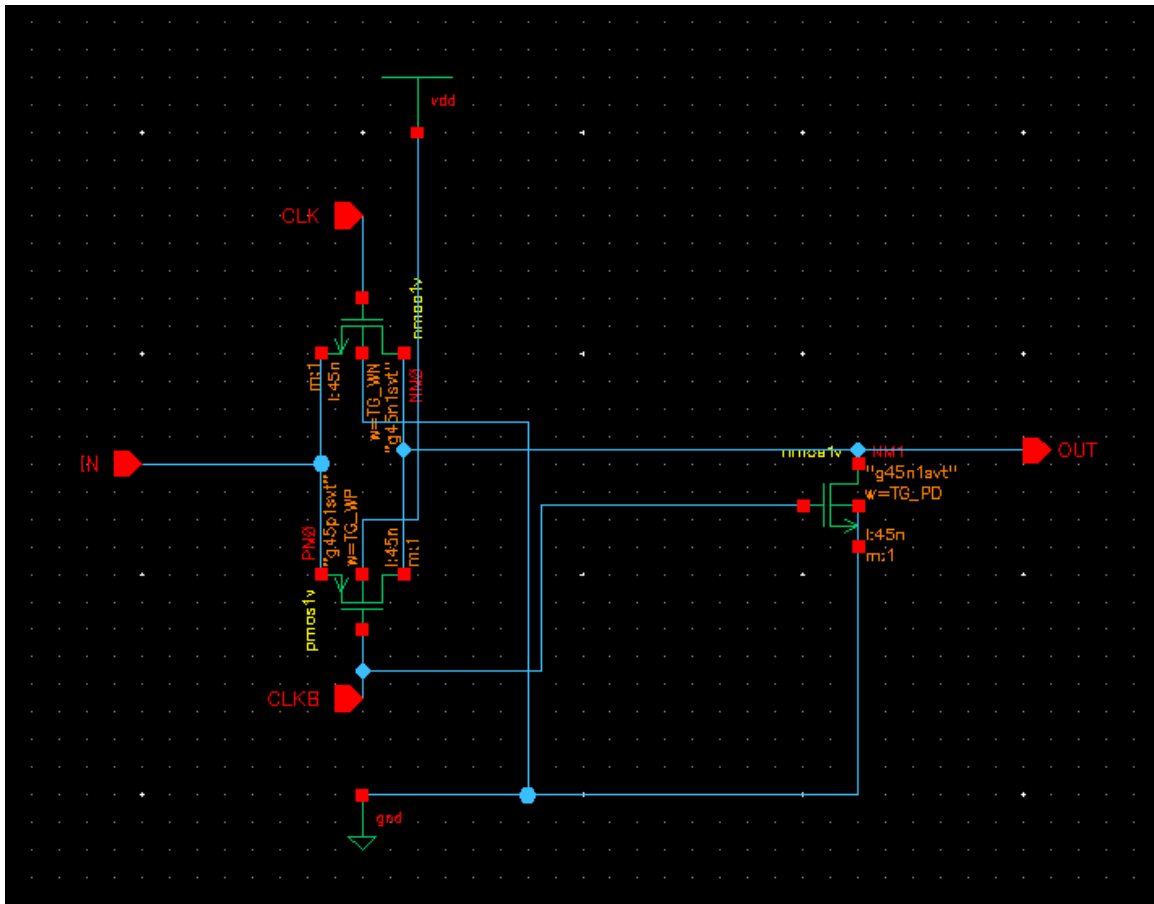


Figure 13. Transmission gate with CLKB activated pull down NMOS. Synchronizes SRAM periphery signals and pulls down node during precharge to avoid corrupting data.

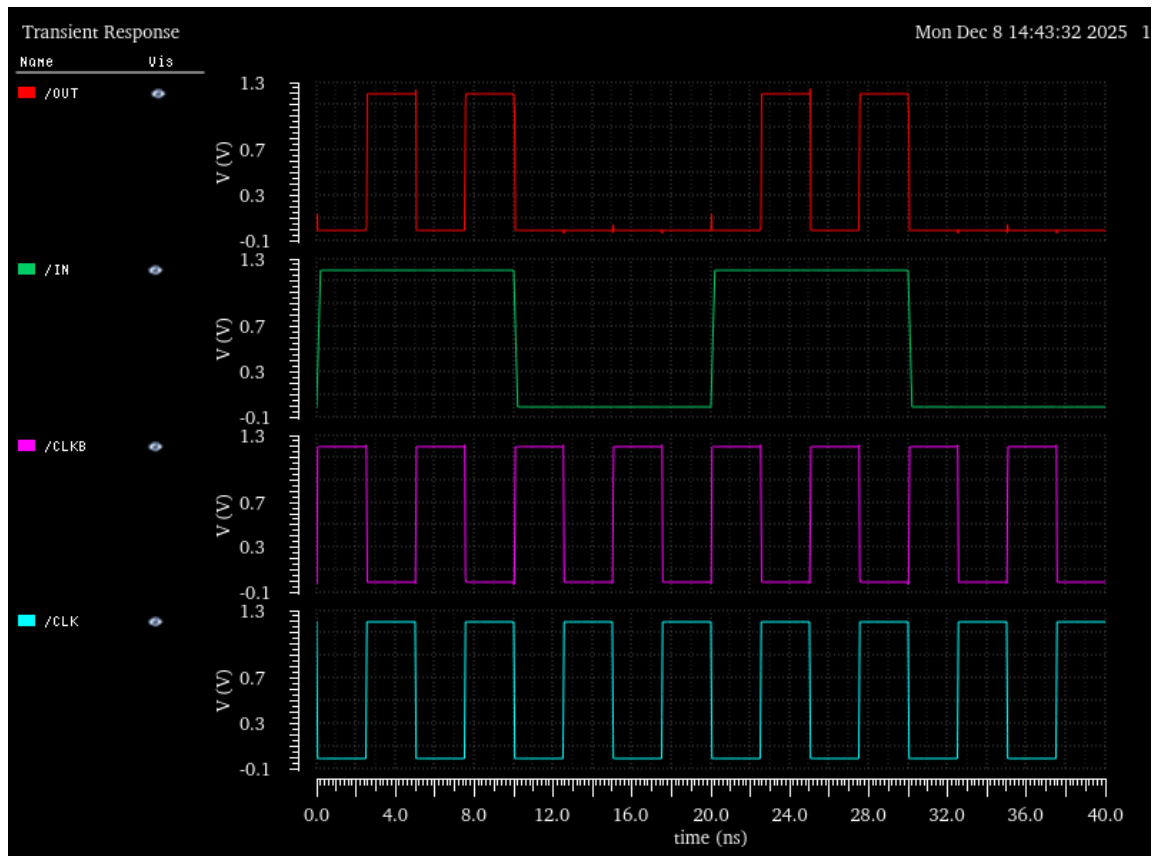


Figure 14. Test output waveforms for TG PD cell. Output follows input on CLK high and gets pulled down to GND on CLKB high.

Transistor Sizing Strategy

The baseline design uses intentional sizing to balance competing requirements:

Transistor Type	Width	Rationale
Pull-up PMOS	1× minimum (120 nm)	Weak feedback during write improves writeability
Access NMOS	2× minimum (240 nm)	Sufficient drive for read/write without excessive loading
Pull-down NMOS	8× minimum (960 nm)	Strong pull-down prevents read disturb, improves SNM

This sizing reflects the classic SRAM tradeoff: strong pull-down and weak pull-up favor read stability at the expense of write margin.

Baseline Test Results

Complete functional verification performed across all 16 rows:

Test Sequence per Row:

Write 0 → Read 0 → Read 0 → Write 1 → Read 1 → Read 1 →

Write 1 → Read 1 → Read 1 → Write 0 → Read 0 → Read 0

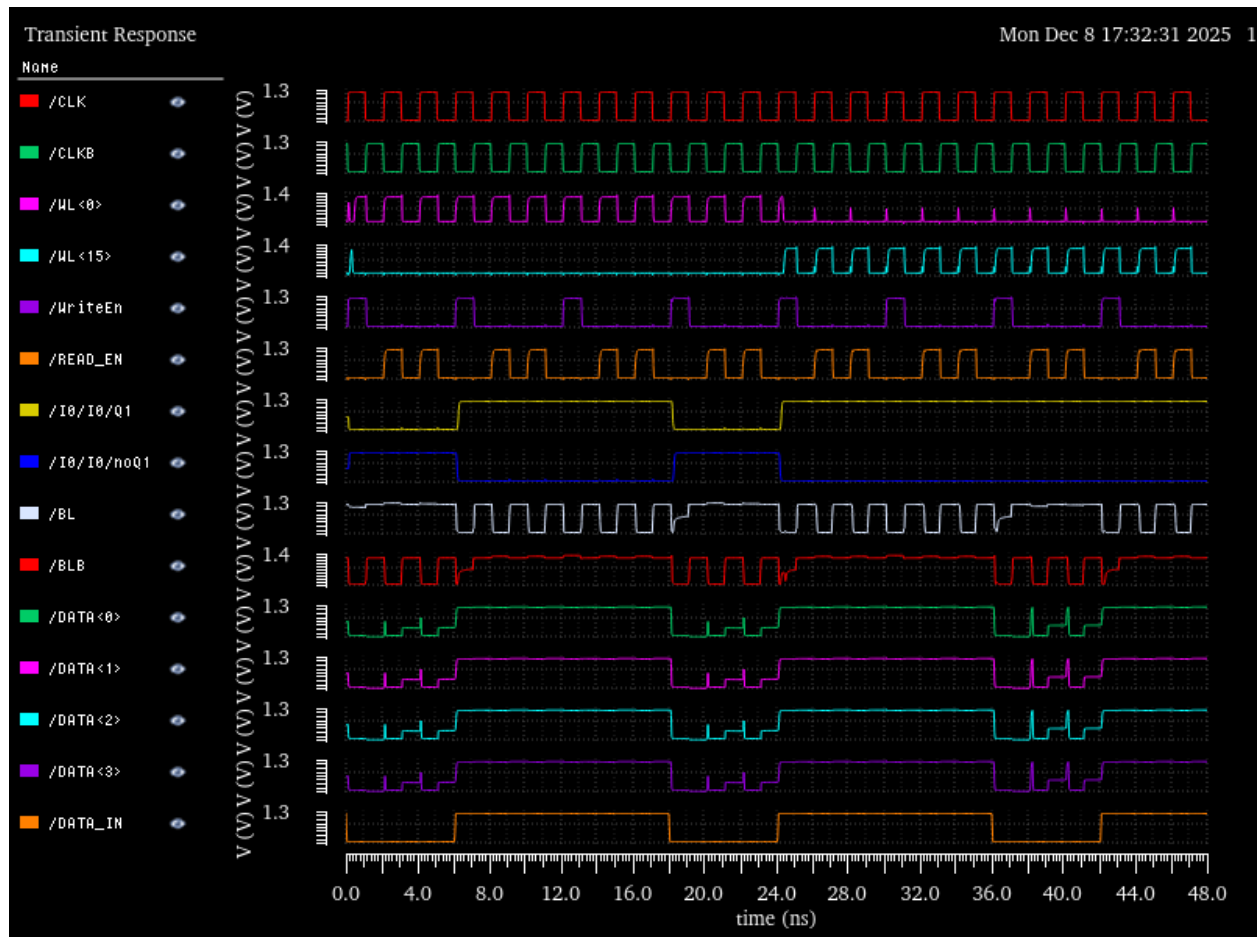


Figure 15. Test output waveforms for the full 64 bit SRAM array. Write and read operations are performed on WL 0 for the first half of the test and WL 15 for the second half.

Key Observations:

- Only one word line active during CLK high ✓
- All word lines grounded during CLKB high ✓
- Bitlines fully precharged every cycle ✓
- Writes occur only when WL and WE both high ✓
- Reads occur only when Read Enable active ✓

3. Optimized Design

Key Optimizations

The optimized design introduces three major improvements:

1. VDD/2 Bitline Precharge → Reduces dynamic power and improves read disturb margin
2. OTA-Based Differential Sense Amplifier → Faster, more sensitive readout
3. Minimum-Sized Memory Cells → Reduced area enabled by improved read margin

Architecture Changes

1. VDD/2 Precharge Network

Replaces full-VDD precharge with mid-supply precharging:

Circuit Implementation:

- Diode-connected PMOS/NMOS stack creates VDD/2 reference ($\sim 0.685V$)
- Inverter stabilizes and buffers reference voltage
- NMOS pass transistors controlled by CLKB apply VDD/2 to bitlines

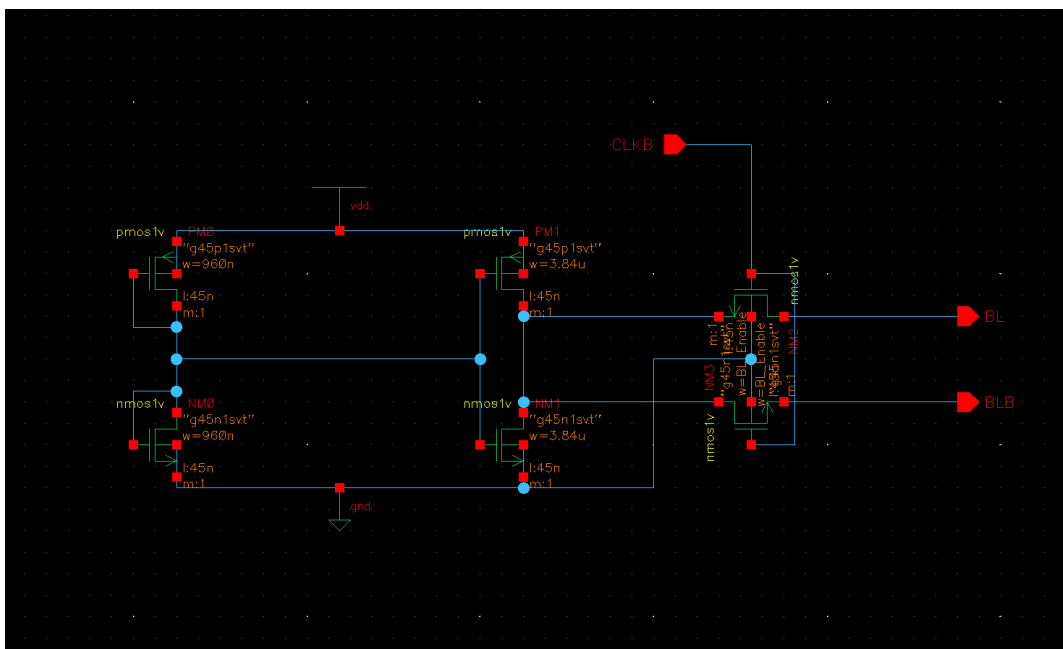


Figure 16. VDD/2 precharge circuit. Stacked diode connected PMOS/NMOS create a half VDD reference voltage, which the inverter holds at its output. Transmission gates pass along the signal to the bitlines.

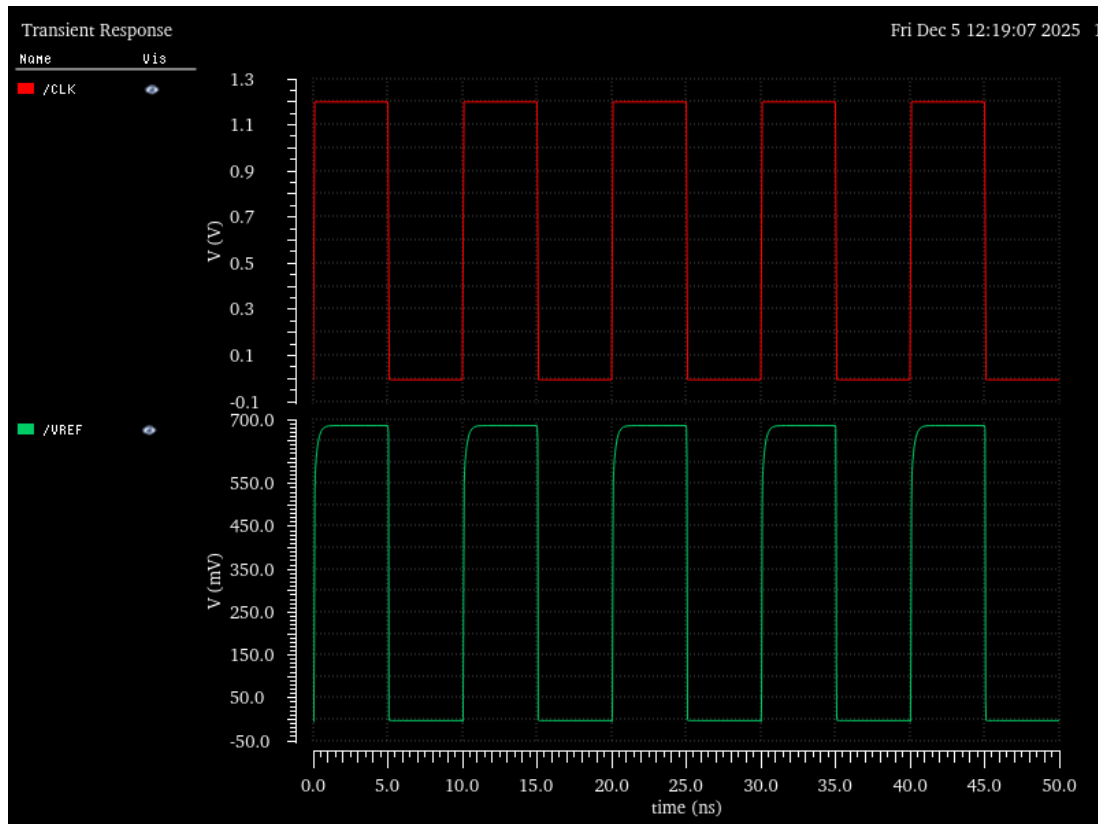


Figure 17. The circuit outputs a steady reference voltage of 0.685V when the input clock is high.

Advantages:

- Reduced Dynamic Power: Bitline swing reduced from VDD to VDD/2
- Improved Read Margin: Internal storage nodes see smaller voltage difference during reads, reducing read disturb risk
- Faster Differential Development: Smaller initial voltage allows faster read sensing
- Better Writeability: Write driver still provides full differential voltage

2. OTA-Based Differential Sense Amplifier

Replaces tri-state buffers with operational transconductance amplifier (OTA):

Circuit Topology:

- NMOS differential input pair connected to BL and BLB
- PMOS active loads for differential-to-single-ended conversion
- Tail current source controlled by Read Enable

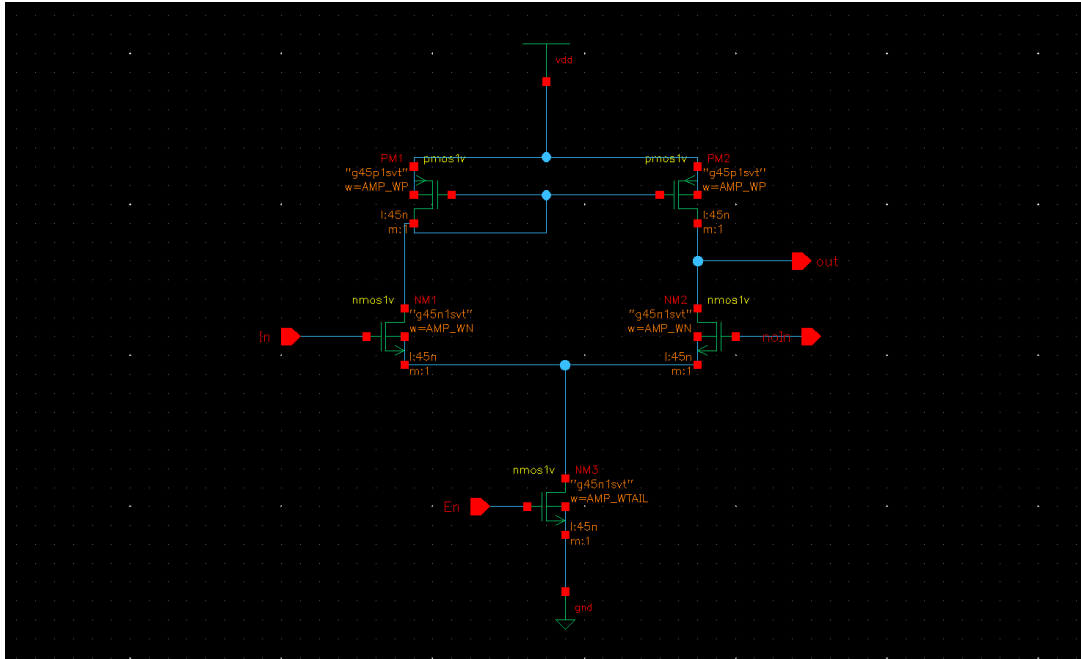


Figure 18. The sense amplifier is an operational transconductance amplifier, providing a single ended output from a differential input.

Advantages:

- High Sensitivity: Amplifies small BL-BLB differences to full logic swing
- Faster Readout: Active amplification faster than passive resistive sensing
- Low Power: Zero static current when Read Enable is low
- Improved Noise Immunity: Differential sensing rejects common-mode noise

3. Minimum-Sized 6T Cells

All transistors in memory cells reduced to minimum width (120 nm):

Why This Works:

- VDD/2 precharge reduces read disturb stress on storage nodes
- Smaller voltage difference between bitlines and internal nodes during reads
- Write driver still provides adequate overdrive for flipping cell state
- Leakage and dynamic power reduced due to smaller devices

Area Reduction: $7.6 \mu\text{m}^2 \rightarrow 2.07 \mu\text{m}^2$ per array (73% reduction)

Optimized Array Architecture

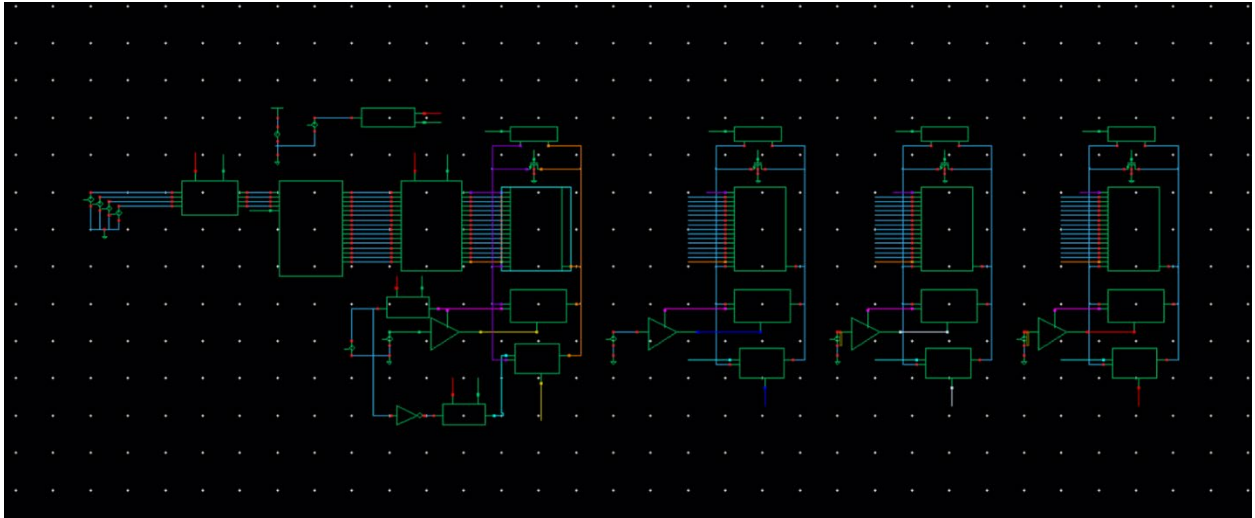


Figure 19. Optimized 64 bit 6T SRAM cell. Includes VDD/2 precharge, sense amp, and min size sram cells.

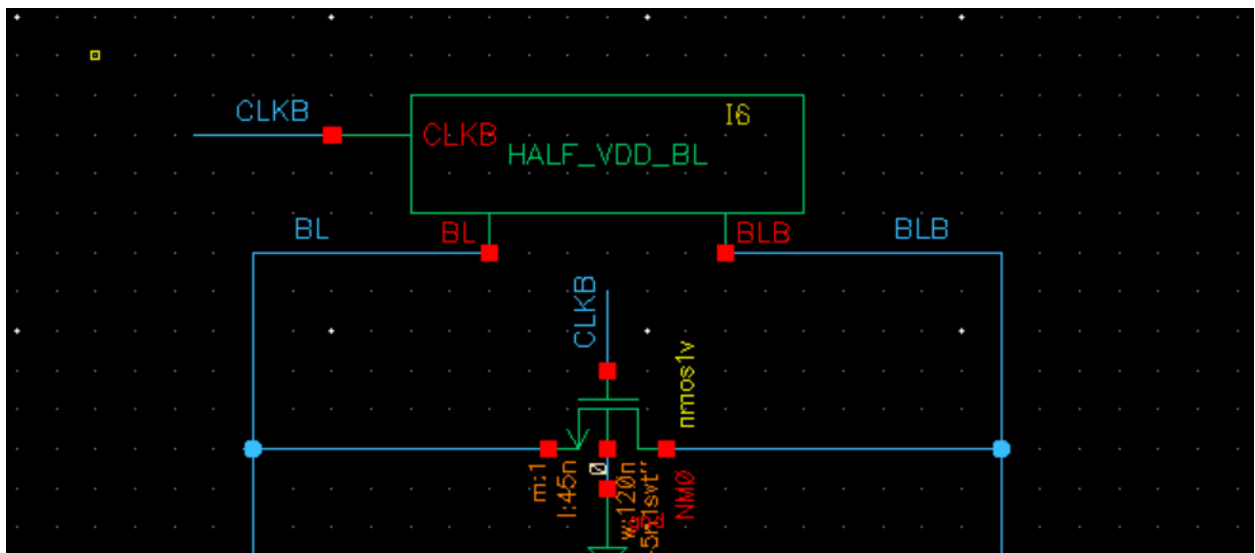


Figure 20. VDD/2 precharge module connected to each bitline and controlled by CLKB.

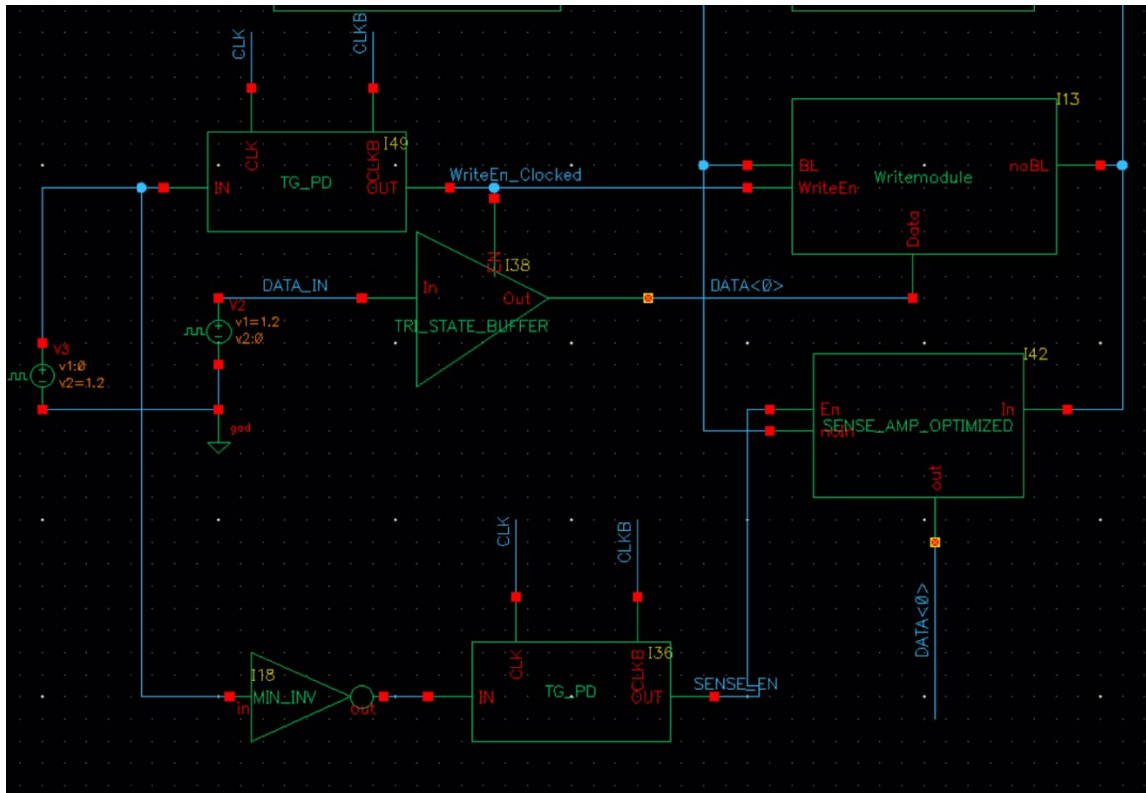


Figure 21. Read/Write circuitry for the optimized array. Replaces the tri state inverters attached to the bitline with the sense amp.

Optimized Test Results

Same functional test sequence as baseline design:

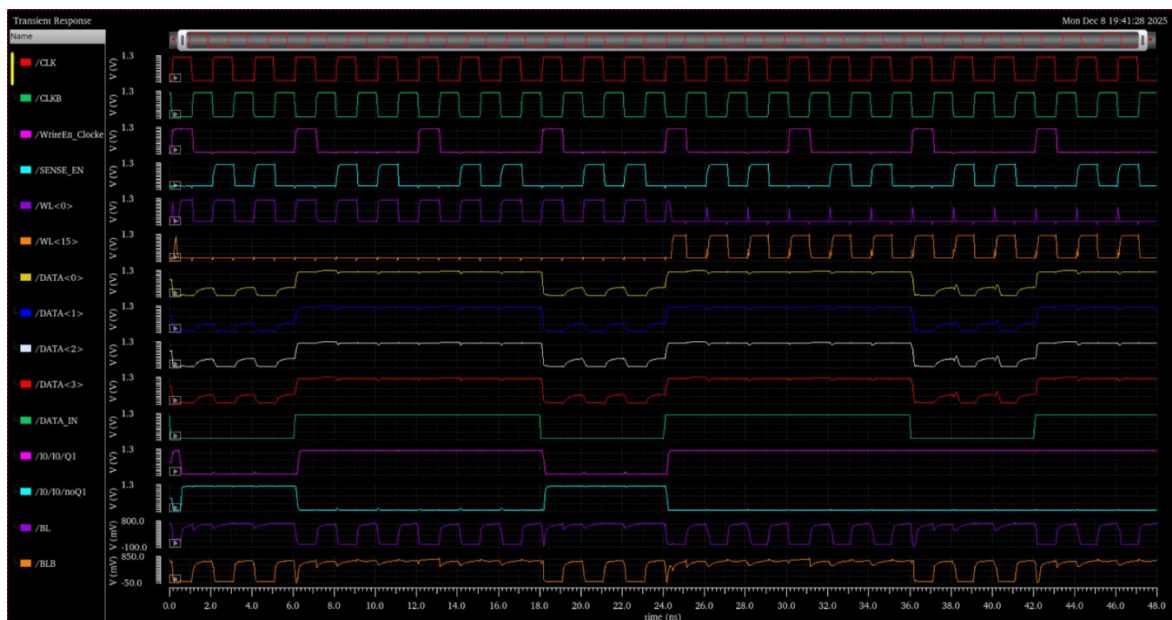


Figure 22. Test output waveforms for optimized array. Bitlines are charged to near VDD/2.

Key Observations:

- Bitlines precharge to $V_{DD}/2$ ($\sim 0.685V$) instead of V_{DD} ✓
 - All read/write operations function correctly with minimum-sized cells ✓
 - Sense amplifiers provide clean full-swing outputs ✓
 - Proper timing maintained through non-overlapping clock phases ✓
-

4. Performance Comparison

Measurement Methodology

Access Time Definitions

Write Access Time (WA):

- Measured from 50% of word line swing to 50% of storage node Q swing
- Separate measurements for high-to-low (WA_{HL}) and low-to-high (WA_{LH}) transitions

Read Access Time (RA):

- Measured from 50% of word line swing to 50% of data bus swing
- Data bus forced to opposite value before read to measure worst-case delay
- Separate measurements for reading '0' (RA_0) and '1' (RA_1)

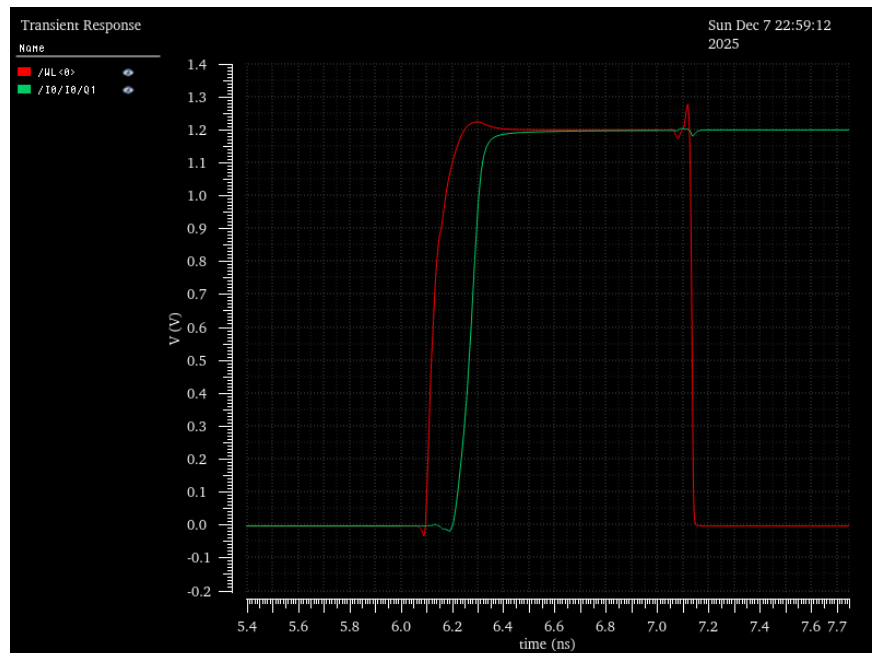


Figure 23. Measurement of the write access time from low to high in the baseline SRAM array

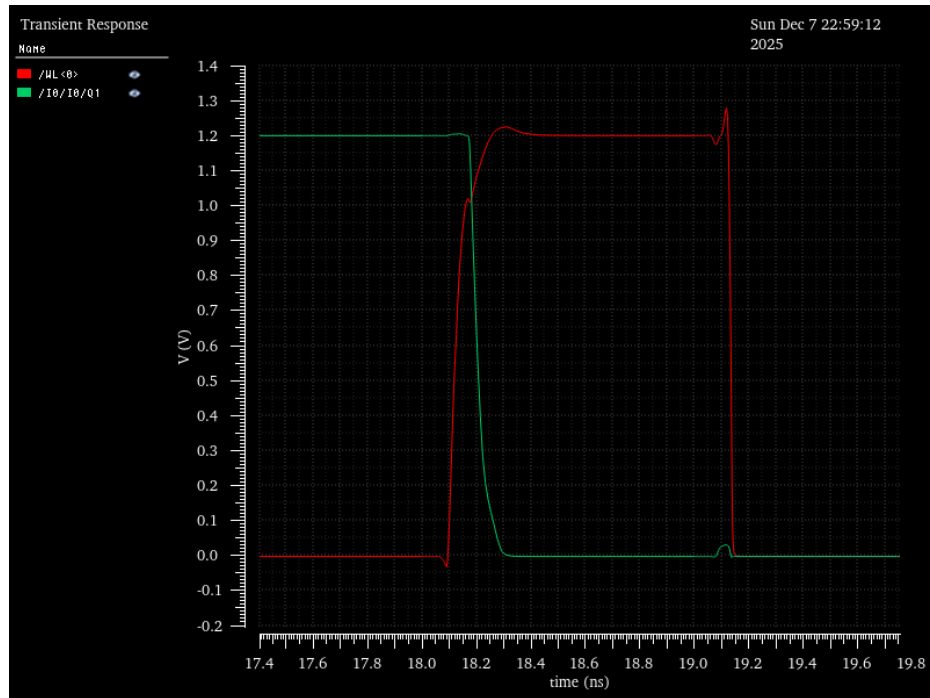


Figure 24. Measurement of the write access time from high to low in the baseline SRAM array

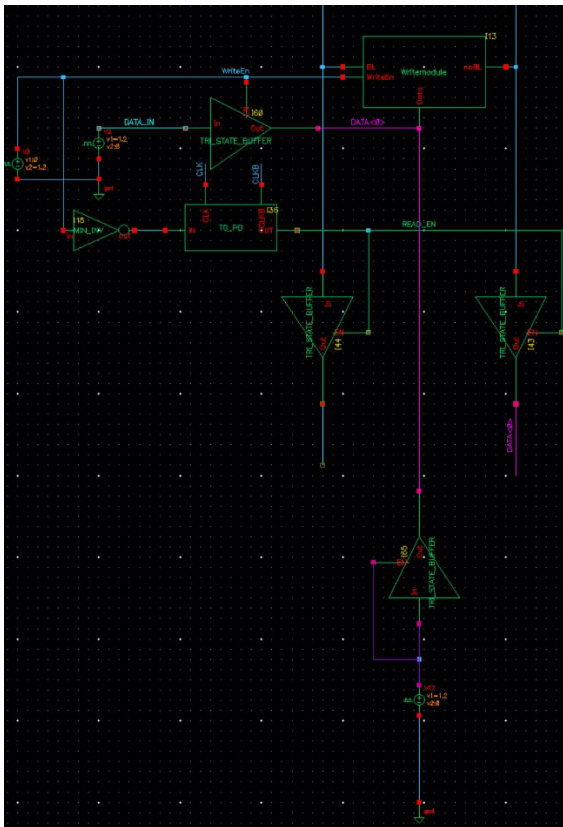


Figure 25. Peripheral circuitry to force a '1' to the data bus

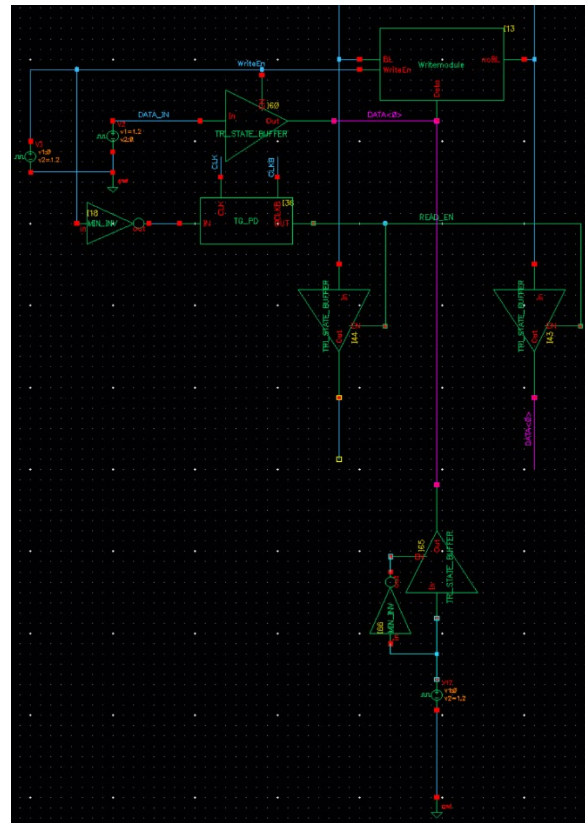


Figure 26. Peripheral circuitry to force a '0' to the data bus

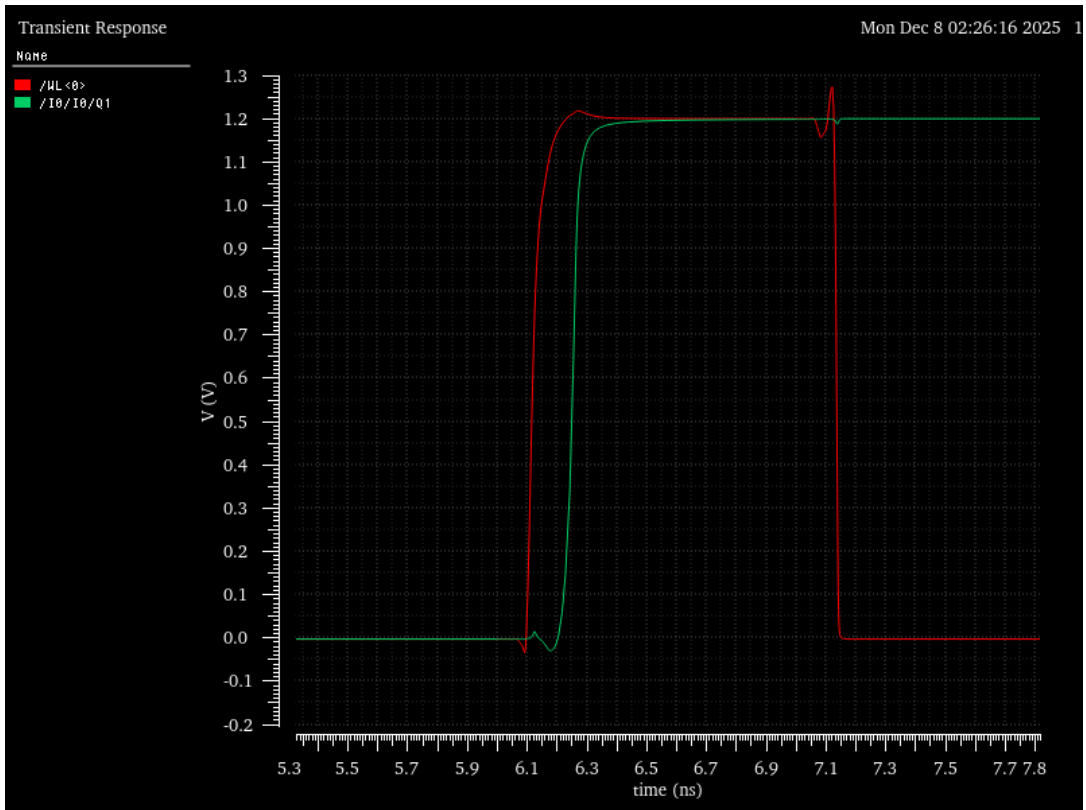


Figure 27. Measurement of the write access time from low to high in the optimized SRAM array

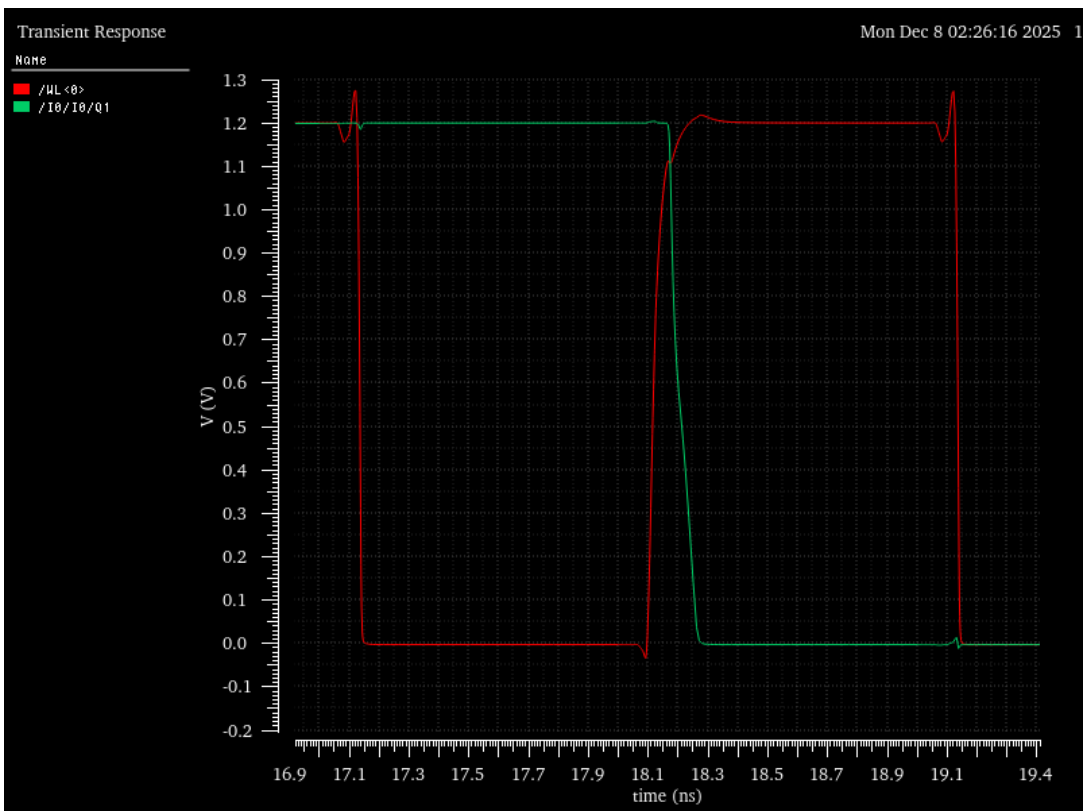


Figure 28. Measurement of the write access time from high to low in the optimized SRAM array

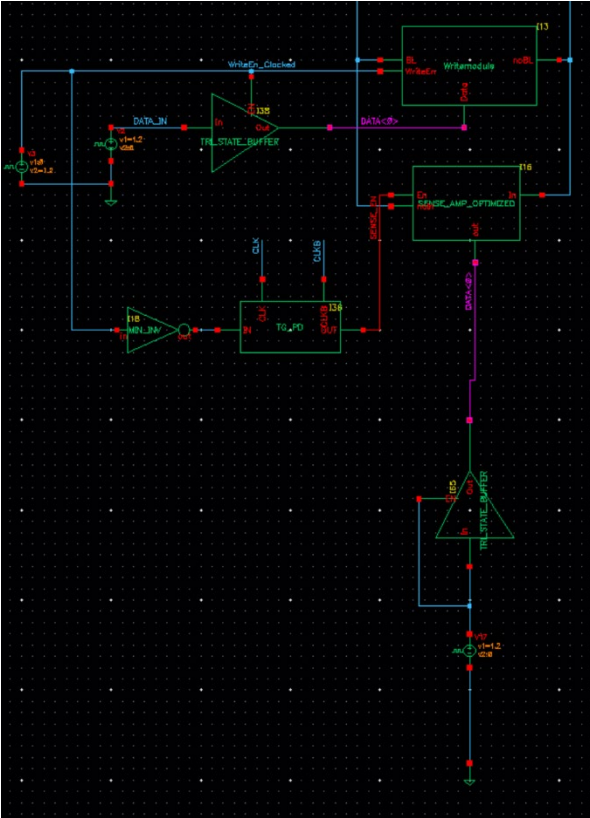


Figure 29. Peripheral circuitry to force a '1' to the data bus

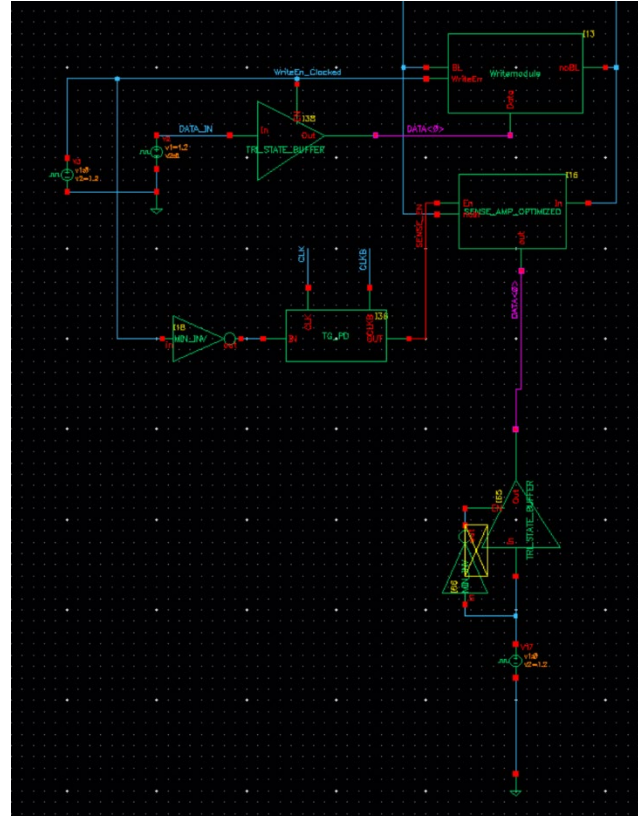


Figure 30. Peripheral circuitry to force a '0' to the data bus

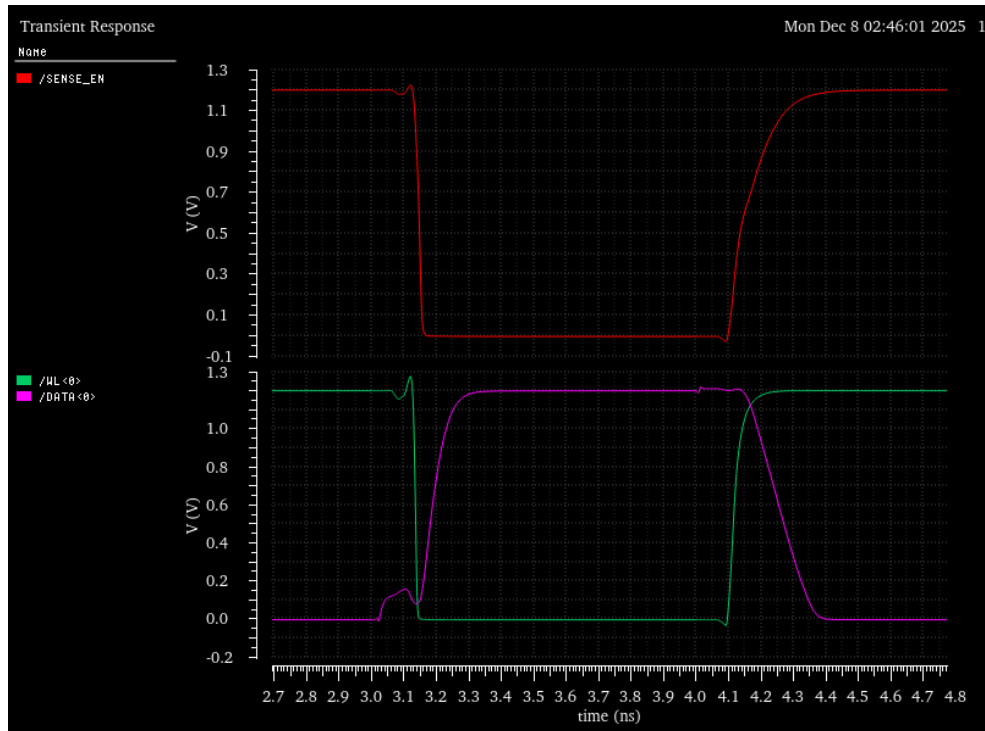


Figure 31. Measurement of the read access time for a '0' in the optimized SRAM array

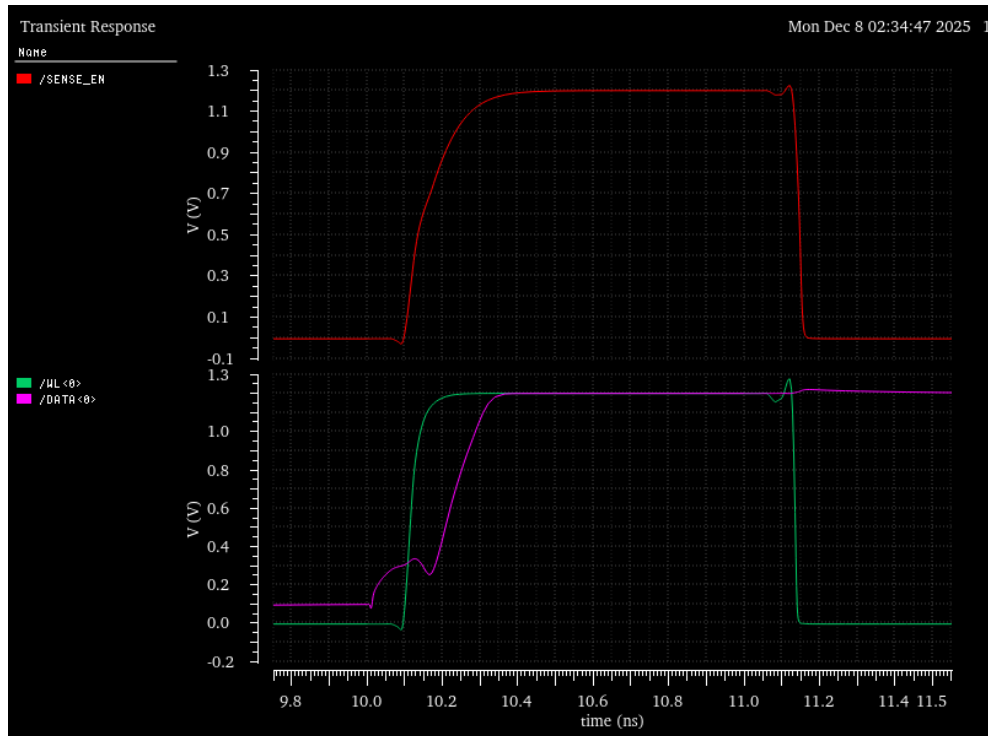


Figure 32. Measurement of the read access time for a '1' in the optimized SRAM array

Maximum Operating Frequency

The maximum switching frequency is not simply the inverse of the worst-case access delay due to setup and hold time requirements. To find the minimum clock period:

1. Start with worst-case access delay as initial period estimate
2. Perform parametric sweep of clock periods
3. Verify storage node Q correctly follows data input for both rising and falling edges
4. Confirm data bus output is valid at end of clock cycle when Read Enable is asserted
5. Select shortest period meeting both criteria

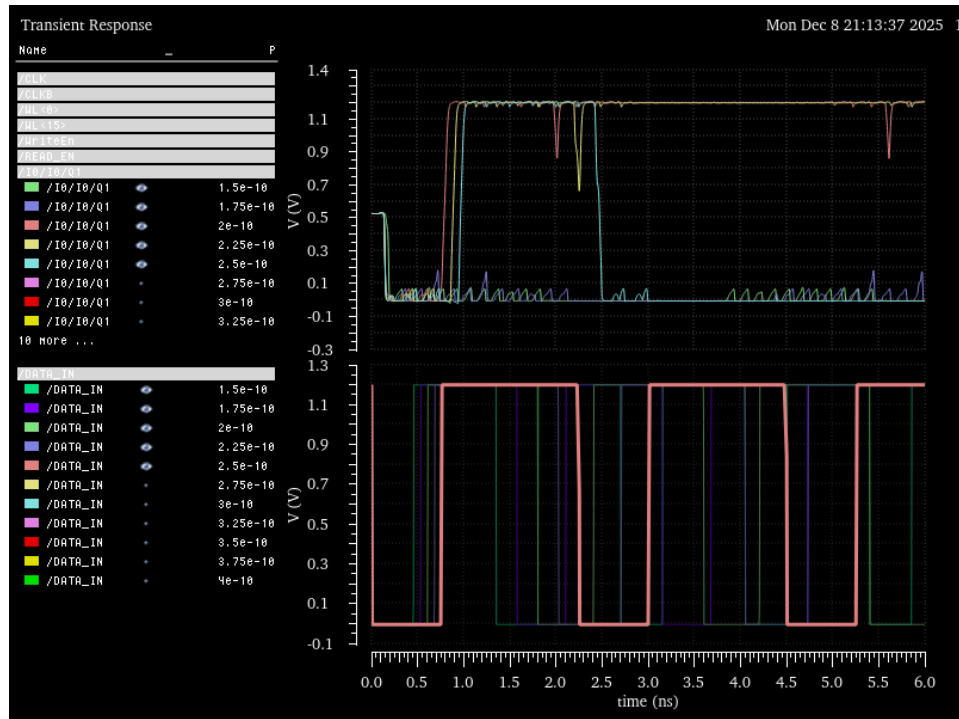


Figure 33. A parametric sweep of clock periods for the baseline design. Since the period of 250 ps is the first whose data storage node Q follows the input data, it is further investigated as the potential minimum period.

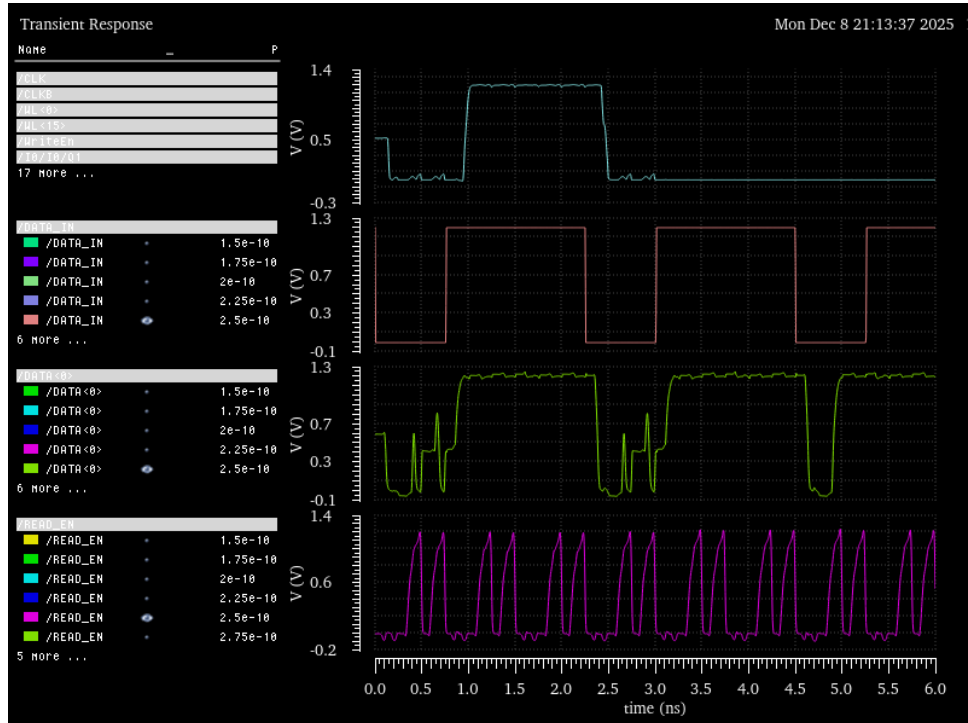


Figure 34. Upon comparing the data bus response to the read enable signal, it is clear that the data does return to the correct output at the very end of the clock cycle, signifying that this period yields correct operation.

Power Measurement

Average power consumption measured during continuous write operations:

- Array written with alternating all 1s and all 0s
- Four full cycles of each pattern
- Operating at maximum switching frequency
- Power calculated by integrating supply current over time

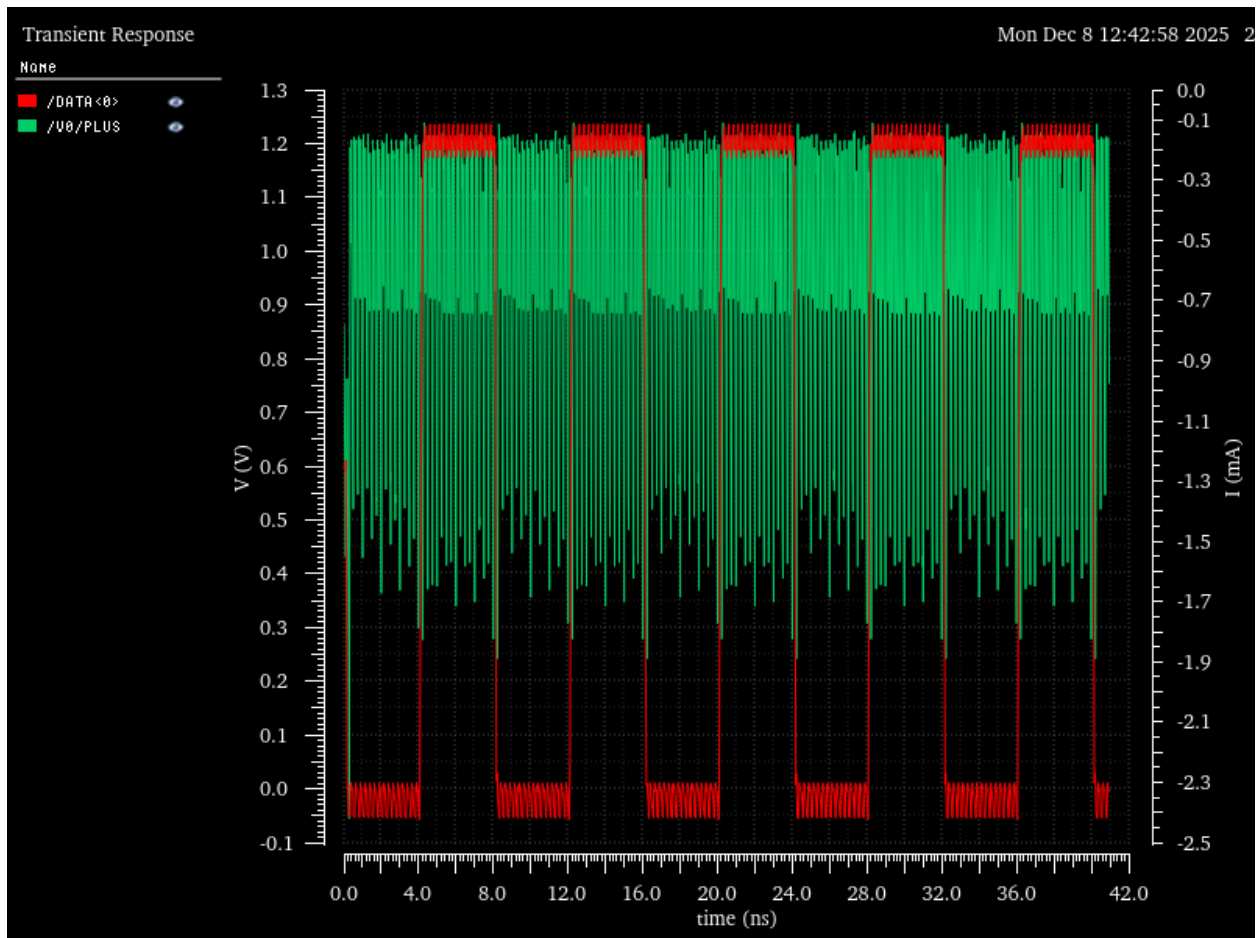


Figure 35. Current draw during the successive write operations performed on the baseline SRAM cell.

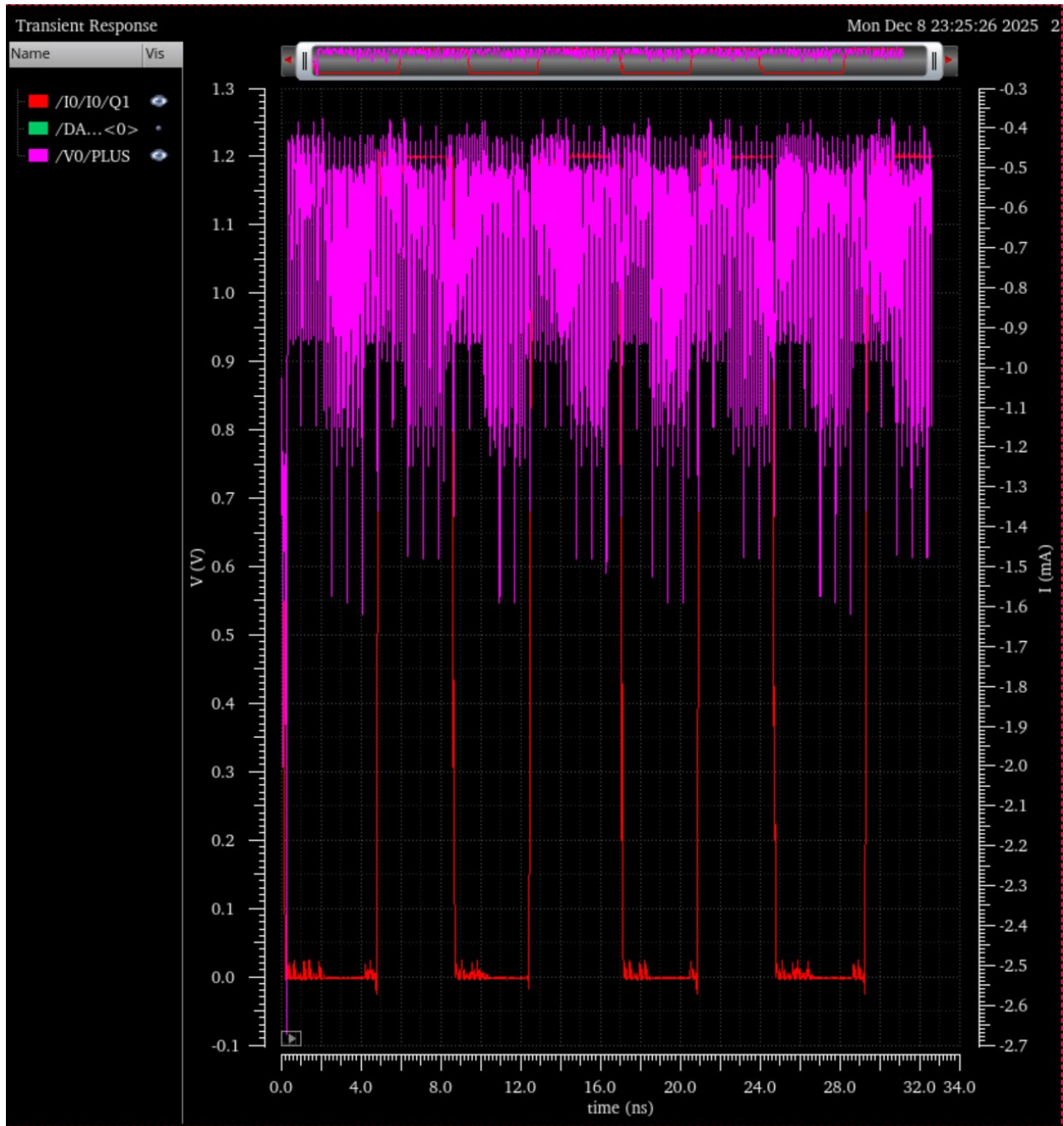


Figure 36. Current draw during the successive write operations performed on the SRAM cell.

Performance Metrics

Access Time Comparison

Design	WA _{HL} (ps)	WA _{LH} (ps)	RA ₀ (ps)	RA ₁ (ps)	Max Delay (ps)
Baseline	78.4	148	89.5	54.7	148
Optimized	81.7	134.4	57.2	121.5	134.4

- Optimized design shows 9% faster worst-case access time
- Write LH delay reduced despite minimum-sized devices
- Read delays vary between designs due to different sensing mechanisms

Frequency and Power Comparison

Design	Min Period (ps)	Max Frequency (GHz)	Power (mW)
Baseline	250	4.0	0.713
Optimized	255	3.92	0.805

- Similar maximum frequencies (~4 GHz)
- Optimized design shows 13% higher power due to sense amplifier biasing current
- Power increase acceptable given substantial area reduction

Area Comparison

Design	Cell Width (nm)	Array Area (μm^2)	Area Reduction
Baseline	PU: 120, ACC: 240, PD: 960	7.6	—
Optimized	PU: 120, ACC: 120, PD: 120	2.07	73%

Calculation methodology:

- Area = 16 rows $\times$ 4 columns $\times$ $[(W_{\text{PU}} \times 2) + (W_{\text{ACC}} \times 2) + (W_{\text{PD}} \times 2)] \times 45\text{nm}$ minimum length
- Peripheral circuit area excluded (assumed negligible compared to array)

5. Key Results

Figure of Merit (FOM)

Definition:

$$\text{FOM} = \text{Area} \times \text{Power} \times \text{Delay}$$

This metric captures the three-way tradeoff between:

- Area: Silicon cost and integration density
- Power: Energy efficiency and thermal management
- Delay: Speed and throughput

Final Comparison

Design	Area (μm^2)	Power (mW)	Delay (ps)	FOM ($\mu\text{m}^2 \cdot \text{mW} \cdot \text{ps}$)
Baseline	7.6	0.713	250	1355
Optimized	2.07	0.805	255	425
Improvement	73% ↓	13% ↑	2% ↑	69% ↓

Key Takeaways

1. Area Dominates FOM: The 73% area reduction from minimum-sized cells outweighs the modest power increase and negligible delay increase
2. VDD/2 Precharge Enables Aggressive Scaling: By reducing read disturb stress, half-VDD precharging allows minimum-sized devices while maintaining stability
3. Sense Amplifier Trade-off: OTA-based sensing provides superior sensitivity and speed but increases static power consumption during reads
4. Co-Optimization is Critical: Peripheral circuits and memory cell sizing must be optimized together to achieve best overall performance
5. 3× FOM Improvement: The optimized design demonstrates that careful architectural choices and circuit-level optimization can dramatically improve memory system efficiency

6. Design Files

This repository contains:

- Full Project Report (Gannon_Sander_Final_Project.pdf): Comprehensive documentation with detailed schematics, test results, and analysis
- Cadence Virtuoso Schematics: All circuit schematics for baseline and optimized designs
- Simulation Testbenches: Complete test setups for functional verification and performance measurement
- Measurement Scripts: Parametric sweep configurations and power analysis setups

Note: All designs were implemented using a proprietary Cadence Virtuoso Process Development Kit (PDK). Simulation files are provided for reference but may not be directly usable without access to the specific PDK.

7. Full Documentation

For complete details including:

- Detailed circuit schematics for all peripheral modules
- Comprehensive timing diagrams and waveforms
- Step-by-step measurement procedures
- Full mathematical derivations of performance metrics
- Extended discussion of design tradeoffs

Please refer to the complete project report: [Gannon_Sander_Final_Project.pdf](#)

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This project was completed as part of ESE 5700: Digital Integrated Circuits and VLSI at the University of Pennsylvania, Fall 2025.